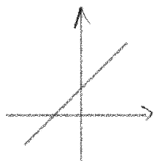


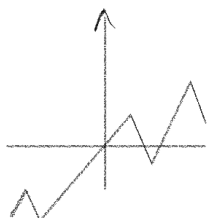
Last time

Ex. Show that there does not exist a continuous function that achieves every value exactly twice.

• Exactly once



• 3 times



Pf. Suppose for a contradiction that such a function existed.

Then in particular, $\exists a, b \in \mathbb{R}$ s.t.

$$f(a) = f(b) = 0 \quad (\text{and } f(x) \neq 0 \forall x \neq a, b)$$

WLOG assume that $f(y) > 0 \forall y \in (a, b)$.

This is true by IVT b/c. if not $f(y_0) < 0$ for some $y_0 \in (a, b)$.

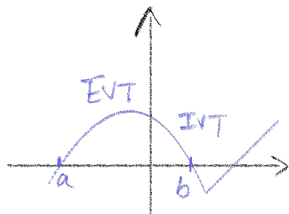
$$f(y_1) > 0 \text{ for some } y_1 \in (a, b).$$

Then $f(y_0) < f(y_1)$ and f is continuous

$\exists z \in (y_0, y_1)$ s.t. $f(z) = 0$, $z \neq a, b$, contradiction

Now, by EVT on $[a, b]$,

$$\exists x \in [a, b] \text{ s.t. } \underset{M}{f(x)} \geq f(y)$$



Since every value is assumed to be achieved twice, $\exists x' \in \mathbb{R}$, $x' \neq x$, $f(x') = M$.

Case ① $x' \notin [a, b]$

Let $0 < N < M$

By IVT, ② Since $f(a) < N < f(x)$

$\exists x_1 \in (a, x)$ s.t. $f(x_1) = N$

⑥ Since $f(b) < N < f(x)$

$\exists x_2 \in (x, b)$ s.t. $f(x_2) = N$.

⑦ WOLOG say $x' < a$

Since $f(a) < N < f(x)$
 $\quad \quad \quad \underset{0}{\parallel} \quad \quad \quad \underset{M}{\parallel}$

$\exists x_3 \in (x', a)$ s.t. $f(x_3) = N$

N is achieved by x_1, x_2, x_3 , contradiction.

If $x' \neq x \in [a, b]$

(i.e. M is achieved twice in $[a, b]$ by x, x' (WOLOG $x < x'$)).

① By EVT on $[x, x']$, a min value $N \neq M$ is achieved by $x_1 \in (x, x')$.

⑥⑦. Since $f(a) < N < f(x)$

$f(x') > N > f(b)$.

$\exists x_2 \in (a, x) \Rightarrow$ s.t. $f(x_2) = f(x_3) = N$

$x_3 \in (x', b)$

By IVT. contradiction



Ex. Let f, g be cts. functions on $[0, 1]$

$$\text{define } \|f\| := \max_{x \in [0, 1]} \{|f(x)|\}$$

norm of f

By EVT, $\|f\|$ is well-defined

Show that $\|f+g\| \leq \|f\| + \|g\|$.

Note. If $f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \in \mathbb{R} \end{cases} \Rightarrow \|f\| = 1$

$$g(x) = \begin{cases} 0, & x \in \mathbb{Q} \\ 1, & x \in \mathbb{R} \end{cases} \Rightarrow \|g\| = 1$$

$$\|f+g\| = 1 \leq \|f\| + \|g\|.$$

Pf. By EVT, $x \in [0, 1]$ s.t. $(f+g)(x) \geq (f+g)(y) \forall y \in [0, 1]$

(EVT) $x_f, x_g \in [0, 1]$ s.t. $f(x_f), g(x_g)$ are max.

$$\|f+g\| \stackrel{\text{EVT}}{=} |(f+g)(x)|$$

$$\stackrel{\text{def}}{=} |f(x) + g(x)|$$

$$\stackrel{\Delta \text{ineq.}}{\leq} |f(x)| + |g(x)|$$

$$\stackrel{\text{EVT}}{\leq} |f(x_f)| + |g(x_g)|$$

$$\stackrel{\text{def}}{=} \|f\| + \|g\|$$

Recall that (P1) - (P12) are satisfied by both $\mathbb{Q}$ and $\mathbb{R}$ but these thms are false for $\mathbb{Q}$

IVT. If $f(a) < 0 < f(b)$ & f cts.

then $\exists c \in (a, b)$ s.t. $f(c) = 0$

Look at $f(x) = x^2 - 2$

- cts. at $x \in \mathbb{Q}$ b/c. cts. $\forall x \in \mathbb{R}$

- let $a = 0$ ($f(a) = -2 < 0$)

$b = 2$ ($f(b) = 2 > 0$)

let $\nexists c \in (0, 2) \cap \mathbb{Q}$ s.t. $f(c) = 0$

so IVT for $\mathbb{Q}$ is false. (need more than (P1)-(P2) to prove this).

EVT. f cts. on $[a, b] \Rightarrow$ max value achieved at some $c \in [a, b]$.

$f(x) = x^3 - x$, local max/min is achieved when $f'(x) = 0$ ($x = \frac{1}{\sqrt{3}}$).

so max/min DNE in $\mathbb{Q}$.

Boundness f cts. on $[a, b] \cap \mathbb{Q}$, then f has an upper bound.

This function is cts. $\forall \mathbb{R}$ except at $\sqrt{2}$.

$\Rightarrow$ cts. at all $\mathbb{Q}$ but not bounded on $[1, 2]$.

